Power Supply 25V 12kA:

proposal report.

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Nomenclature

- AFE: active front end;
- HF-TR: high-frequency transformer;
- PS-FBC: phase-shift full bridge converter;
- LCL: inductor-capacitor-inductor filter;
- $\mathrm{THD_i}$: current total harmonic distortion;
- L_{grid} $\left[\mathrm{H}\right]$: point of common coupling (PCC) inductance and *LCL* output filter inductance;
- C_{Fu} $\left[\mathrm{F}\right]$: *LCL* output filter capacitor;
- R_{CFu} $\left[\Omega\right]$: damping of the LCL output filter capacitor;
- L_{Fu} $\left[\mathrm{H}\right]$: *LCL* input filter inductance;
- C_{Fi} $\left[\mathrm{F}\right]$: dclink capacitor of the AFE stage;
- C_{dc}^{dc1} $\left[\mathrm{F}\right]$: input capacitor of the PS-FBC stage;
- C_{dc}^{dc2} $\left[\mathrm{F}\right]$: output capacitor of the PS-FBC stage;
- L_{dc}^{dc2} $\left[\mathrm{H}\right]$: output inductance of the PS-FBC stage;
- L_{s} $\left[\mathrm{H}\right]$: series inductance of the PS-FBC stage;
- $\vec{u}_{\mathrm{g}}^{\mathrm{abc}}(t)$ $\left[\mathrm{V}\right]$: grid voltage used by AFE for PLL grid synchronization;
- $\vec{i}_{\mathrm{g}}^{\mathrm{abc}}(t)$ $\left[\mathrm{A}\right]$: grid current used for PF and $\mathrm{THD_i}$ evaluation;
- $u_{\mathrm{dc}}^{\mathrm{dc1}}(t)$ $\left[\mathrm{V}\right]$: AFE stage dclink voltage;
- $u_{\mathrm{ac}}^{\mathrm{ac1}}(t)$ $\left[\mathrm{V}\right]$: transformer input voltage;
- $i_{\mathrm{ac}}^{\mathrm{ac1}}(t)$ $\left[\mathrm{A}\right]$: transformer input current;
- $i_{\mathrm{d}}^{\mathrm{dn}}(t)$ $\left[\mathrm{A}\right]$: n-th diode rectifier current;
- $f_{\mathrm{pwm}}^{\mathrm{afe}}$ $\left[\mathrm{Hz}\right]$: AFE switching frequency;
- $f_{\mathrm{pwm}}^{\mathrm{fbc}}$ $\left[\mathrm{Hz}\right]$: PS-FBC switching frequency.

1 Introduction

The system described in this document is designed to deliver a nominal output voltage of 25 V at 12 kA, corresponding to a rated power of 300 kW, from a three-phase 480 V mains supply.

The document is structured as follows. Section 2 summarises the electrical and thermal specifications together with the two operating conditions defined for compliance verification: nominal load cold (**NLC**) and nominal load hot (**NLH**). Section 4 presents the selected topology and justifies the design choices with respect to efficiency, current ripple, and controllability. The converter design — including transformer architecture, switching-stage sizing, and output filter — is detailed in Section 3. Simulation results are shown in Section 5.

2 Power Supply Specification

In this section a list of requirements and constraints which the power supply shall satisfy is reported.

- Overall performance and requirements:
  - Nominal Output Voltage, $V_{\text{out}}^{\text{nom}} = 25\,\text{V}$.
  - Nominal Output Current, $I_{\text{out}}^{\text{nom}} = 12\,\text{kA}$.
  - Nominal Output Power, $P_{\text{out}}^{\text{nom}} = 300\,\text{kW}$.
  - Maximum Output Voltage Ripple, $\Delta v_{\text{out}} \leq 0.25\,\text{V}$ (rms).
  - Current accuracy, $|\tilde{i}|_{\max} \leq 1\,\%$.
  - Voltage accuracy, $|\tilde{v}|_{\max} \leq 0.5\,\%$.
  - Efficiency, $\eta > 90\,\%$.
  - The power supply shall provide galvanic isolation.
  - Maximum grid current distortion, $\text{THD}_i < 5\,\%$.
  - Power Factor, $\text{PF} > 0.95$ over the entire operating range.
- Electrical power source specifications:
  - Grid Voltage, $V_{\text{grid}} = 480\,\text{V}$, operative range $+5\,\%$ / $-10\,\%$.
  - Grid Frequency, $f_{\text{grid}} = 50\,\text{Hz}$ / $60\,\text{Hz}$, $\pm 2\,\%$.
  - Short-Circuit Current at PCC, $I_{\text{cc}} = 15\,\text{kA}$.
- Liquid cooled (de-ionized water not specified).
- Operative working condition as shown in Figure 1.

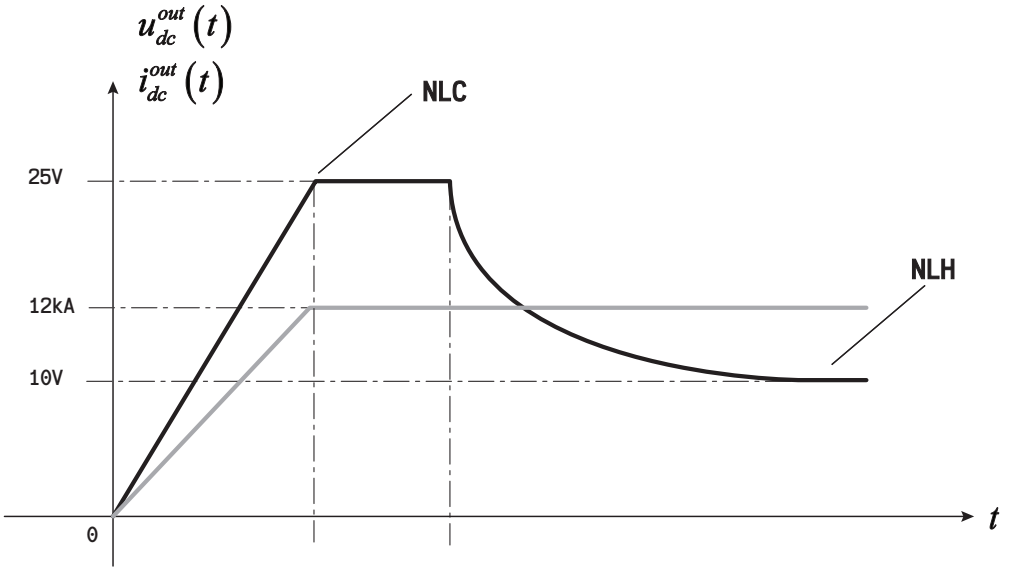


Figure 1: Load current and voltage profile under different operative conditions (NLC and NLH).

3 Power Supply Architecture

The demanding nature of this specification — in particular the combination of very high output current, tight ripple and regulation requirements, and the need for galvanic isolation — calls for a carefully selected power conversion architecture. The proposed solution (as suggested by SECOM srl) consists of two power conversion stages:

- an Active Front End (AFE) rectifier;
- an isolated DC/DC converter based on a Full Bridge Converter (here called PS-FBC, where PS stands for phase-shifted).

As set out in the specifications above, satisfying the requirements of unity-adjacent power factor (PF $>$ 0.95), low grid current distortion ($\mathrm{THD}_i < 5\%$), and galvanic isolation calls for the two-stage solution described hereafter, comprising an AFE stage (see Figure 2) and a PS-FBC stage (see Figure 3).

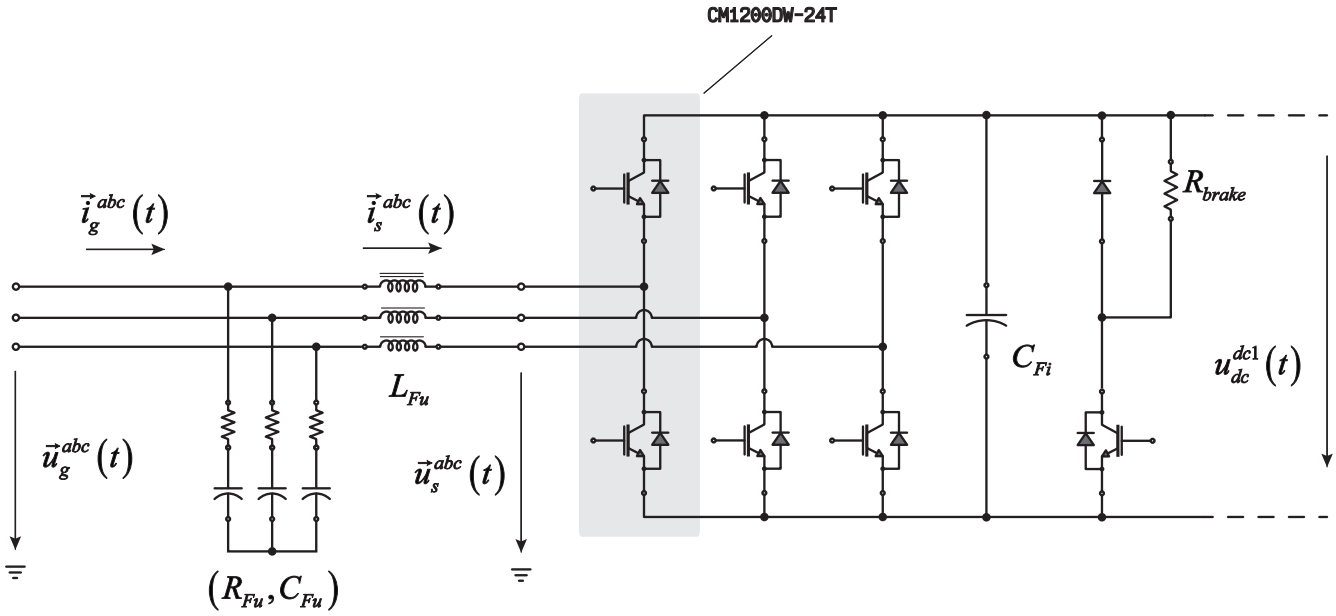


Figure 2: Active Front End stage with LC filter layout.

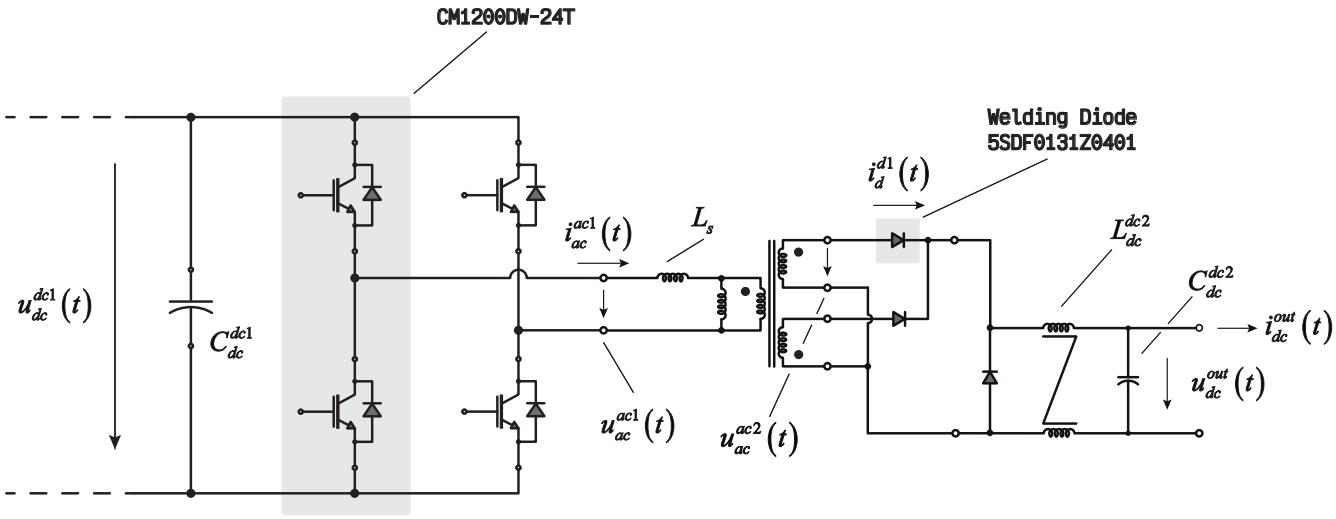


Figure 3: Full Bridge Converter layout.

The source inductance L_{grid}, determined by the grid short-circuit capacity at the point of common coupling (15 kA), represents the dominant upstream impedance seen by the AFE stage.

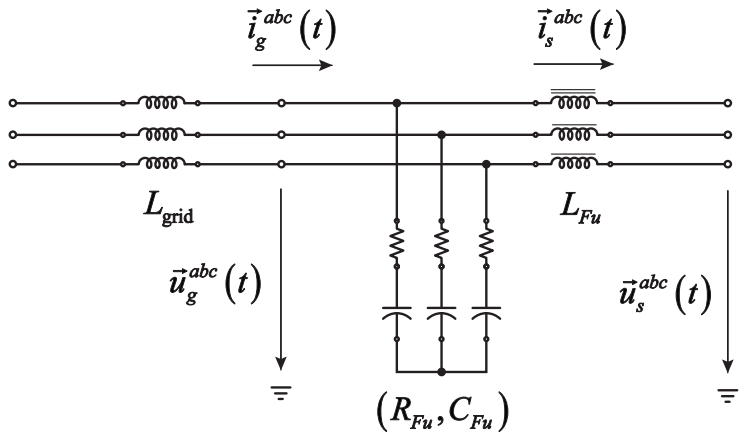


Figure 4: LCL filter built using the point of common coupling (PCC) inductance.

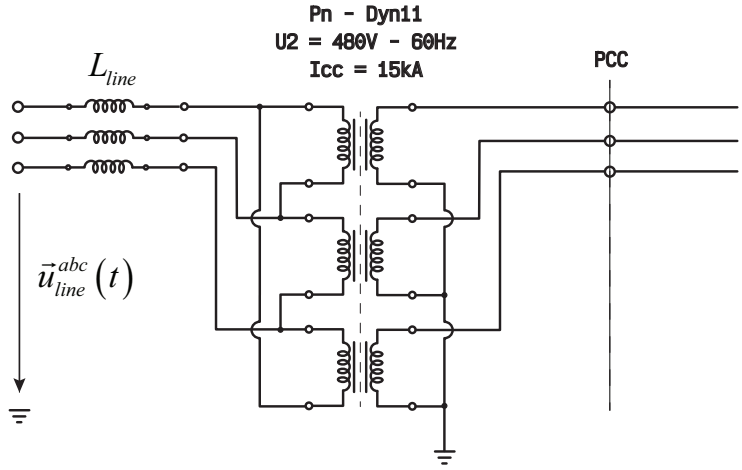


Figure 5: Line representation.

4 Power Supply Design

List of the power supply components and related sizing.
Remark - the design has been proposed by SECOM srl.

Table 1: Component design.

component	Nominal data	Design criteria data	description
AFE L_{Fu}	$L_{\text{Fu}} = 330\,\mu\text{H}$	$I_{\text{nom}} = 450\,\text{A}$ rms	liquid cooled
AFE C_{Fu}	$C_{\text{Fu}} = 200\,\mu\text{F} - 360\,\mu\text{F}$	$U_{\text{nom}} = 285\,\text{V}$ rms, $I_{\text{nom}} = 44\,\text{A}$ rms	film capacitor
AFE R_{CFu}	$R_{\text{CFu}} = 50\,\text{m}\Omega$	$P_{\text{nom}} = 50\,\text{W}$	low inductance resistors
AFE L_{grid}	$L_{\text{grid}} = 50\,\mu\text{H}$		from grid

AFE IGBT module power losses	CM1200DW-24T, $R_g = 1.15\,\Omega$, $f_{\text{pwm}} = 4\,\text{kHz}$	$P_{\text{loss}} = 1200\,\text{W}$ per module under worst condition	liquid cooled
AFE DClink Capacitor	$C_{\text{Fi}} = 2.4\,\text{mF}$	$U_{\text{nom}} = 900\,\text{V}$	film capacitor
FBC Input Capacitor	$C_{\text{dc}}^{\text{dc1}} = 1.2\,\text{mF}$	$U_{\text{nom}} = 900\,\text{V}$	film capacitor
FBC IGBT module power losses	CM1200DW-24T, $t_{dt} = 3\,\mu\text{s}$, $f_{\text{pwm}} = 10\,\text{kHz}$	$P_{\text{loss}} = 1200\,\text{W}$ per module under worst condition	liquid cooled
HF Trafo series inductance	$L_s = 8\,\mu\text{H}$	center tapped transformer, $f_{\text{pwm}} = 10\,\text{kHz}$	
HF Trafo Primary Current	$I_1 = 690\,\text{A}$		RMS value for sizing
HF Trafo Primary Voltage	$U_1 = 680\,\text{V}$		RMS value for sizing
HF Trafo Secondary Current	$I_2 = 8400\,\text{A}$ per system		RMS value for sizing
HF Trafo Secondary Voltage	$U_2 = 35\,\text{V}$ per system		RMS value for sizing
Diode Rectifier Power losses	5SDF-0131Z0401	$P_{\text{loss}} = 6\,\text{kW}$, $I_d^m = 3000\,\text{A}$ per diode	Clamped plate and Snubbers
FBC $L_{\text{dc}}^{\text{dc2}}$	$L_{\text{dc}}^{\text{dc2}} = 0.4\,\mu\text{H}$	$I_{\text{nom}} = 12\,\text{kA}$	in air, maybe from bus bar partial loop inductance
FBC $C_{\text{dc}}^{\text{dc2}}$	$C_{\text{dc}}^{\text{dc2}} = 50\,\text{mF}$	$I_{\text{nom}} = 350\,\text{A}$ rms	film (mandatory)

4.1 General design aspects

Here a list of preliminary assumptions, and design choices:

- Nominal AFE power rate has been selected at $P_{\text{nom}}^{\text{afe}} = 325\,\text{kW}$.
- Nominal AFE stack current has been selected at $I_{\text{nom}}^{\text{afe}} = 435\,\text{A}$ (rms).
- AFE pwm switching frequency has been selected at $f_{\text{pwm}}^{\text{afe}} = 4\,\text{kHz}$.
- PS-FBC pwm switching frequency has been selected at $f_{\text{pwm}}^{\text{fbc}} = 10\,\text{kHz}$ (operative frequency of the transformers).

– The overall architecture, AFE+PS-FBC has been simulated assuming **CM1200DW-24T** ($T_j^{\max} = 175\,^{\circ}\mathrm{C}$) as switching device. Maximum power loss for each **module** remains below than 1.4 kW along the worst operative condition.

Remark - the secondary rectifier stage of the PS-FBC has been implemented using slow-recovery diodes; the Simscape model has been updated accordingly to include a full diode model.

The selected switching frequency of 10 kHz represents a design compromise between volume and sound emission. The choice of 10 kHz allows the use of ferrite core.

4.2 AFE-LCL filter topology

As shown in Figure 4 the active-front-end is interfaced to the grid via an LCL filter (even if the input inductor (L_{Fu}) is mandatory for the conversion). In this context and assuming an $I_{\mathrm{cc}} = 15\,\mathrm{kA}$ which results into a $L_{\mathrm{grid}} \approx 54\,\mu\mathrm{H}$.
Remark - the LCL filter will be composed by an LRC filter (L_{Fu}, C_{Fu}, and R_{CFu}) plus the point of common coupling inductance (L_{grid}). The $R_{\mathrm{CFu}} = 50\,\mathrm{m}\Omega$ is basically used as damping and in general helps to achieve harmonics compliance in the scenario of power generation where, as according to IEEE519, the harmonic limits are reduced by a factor of two.

4.3 AFE DClink

The DClink capacitor has been selected as follows

– Film capacitor DClink bank;

– DClink bank value $C_{Fi} \approx 2.4\,\mathrm{mF}$–$3.6\,\mathrm{mF}$;

– DClink bank rated voltage $U_{C_{\mathrm{Fi}}}^{\mathrm{nom}} \approx 900\,\mathrm{V}$–$1300\,\mathrm{V}$.

4.4 Phase Shift Full bridge Converter topology

Rectifier stage - rectifier stage is implemented using the component 5SDF-0131Z0401. In order to evaluate the effect of the reverse recovery current, diode simscape model has been implemented using the junction charge model.

HF Transformer stage - The series inductance L_s, comprising the transformer leakage inductance L_{lk} and an optional external inductor L_{ext}, plays a central role in the operation of the Phase-Shift Full Bridge Converter (PS-FBC). It stores the energy required to charge and discharge the IGBT output capacitance C_{oss} during the dead-time interval, thereby enabling Zero-Voltage Switching (ZVS). Furthermore, L_s governs the power transferred between primary and secondary according to the phase-shift angle $\varphi(t)$, and provides inherent $\mathrm{d}i/\mathrm{d}t$ limiting during fault conditions.

Transformer has been sized as follows

– centre-tapped transformer with a turns ratio of $n_1/n_2 = n_1/n_3 = 17$;

– $L_s = 8\,\mu\mathrm{H}$ per transformer which represents a trade-off between ZVS range and maximum power delivery;

4.5 AFE and FBC Devices

The primary full-bridge stage as well as the three phase inverter are built using the LV100 module CM12000DW-24T which has a maximum junction temperature $T_j^{\max} = 175\,^{\circ}\mathrm{C}$.

4.6 Overall Efficiency

AFE — An AFE with a rated current of 450 A, a switching frequency of 4 kHz, and a DC-link voltage of 750 V requires liquid cooling. The overall AFE losses are estimated at approximately 5 kW, comprising 1.4 kW in the commutation inductance L_{Fu} and 3.6 kW in the IGBT modules.
FBC — Similar considerations apply to the FBC stage. The full-bridge switching losses are estimated at approximately 3 kW under the worst-case operating condition.
HF-TR — For the HF transformer, a total loss of 5 kW is assumed, accounting for both core and winding losses.
Rectifier — With two diode modules each carrying 6 kA, the total rectifier conduction loss amounts to approximately 12 kW.
Bus Bars — overall bus bar connection power losses has been assumed approximately 2 kW.
The individual loss contributions are summarised below:

- $P_{\mathrm{loss}}^{\mathrm{AFE}} = 3.6\,\mathrm{kW}$;
- $P_{\mathrm{loss}}^{\mathrm{LFu}} = 1.4\,\mathrm{kW}$;
- $P_{\mathrm{loss}}^{\mathrm{FBC}} = 3.0\,\mathrm{kW}$;
- $P_{\mathrm{loss}}^{\mathrm{HF\text{-}TR}} = 5.0\,\mathrm{kW}$;
- $P_{\mathrm{loss}}^{\mathrm{RECT}} = 12.0\,\mathrm{kW}$;
- $P_{\mathrm{loss}}^{\mathrm{BusBars}} = 2.0\,\mathrm{kW}$;

Remark — The total estimated loss is $P_{\mathrm{loss}}^{\mathrm{tot}} \approx 27\,\mathrm{kW}$, corresponding to an overall system efficiency of $\eta = P_{\mathrm{out}}/(P_{\mathrm{out}} + P_{\mathrm{loss}}^{\mathrm{tot}}) = 300/(300+27) \approx 91.7\,\%$, which satisfies the requirement $\eta > 90\,\%$.

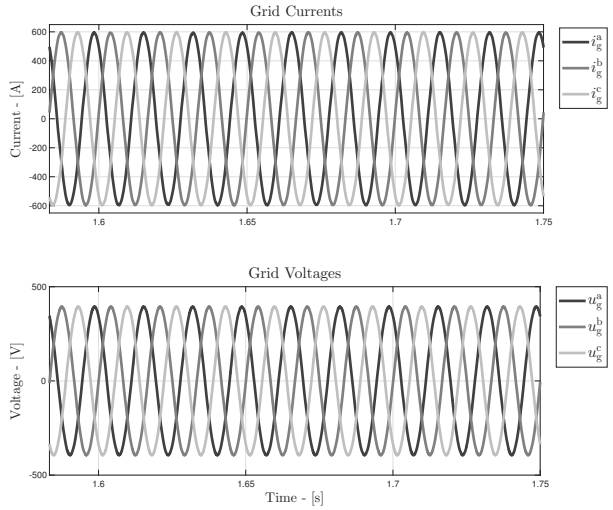
5 Simulation Results

Simulations have been performed at two working condition

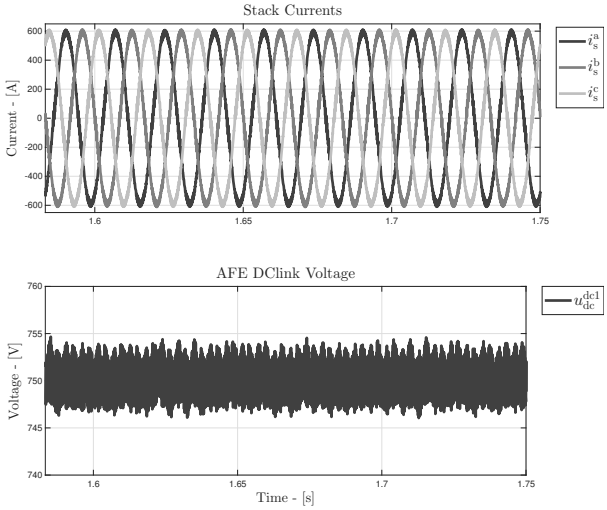
- NLC: $u_{\mathrm{dc}}^{\mathrm{out}} = 25\,\mathrm{V}$ and $i_{\mathrm{dc}}^{\mathrm{out}} = 12\,\mathrm{kA}$;
- NLH: $u_{\mathrm{dc}}^{\mathrm{out}} = 10\,\mathrm{V}$ and $i_{\mathrm{dc}}^{\mathrm{out}} = 12\,\mathrm{kA}$;

5.1 Case scenario NLC

This subsection presents the simulation results for the NLC case. The simulations report the voltages and currents of the main passive and active components, as well as the corresponding losses.

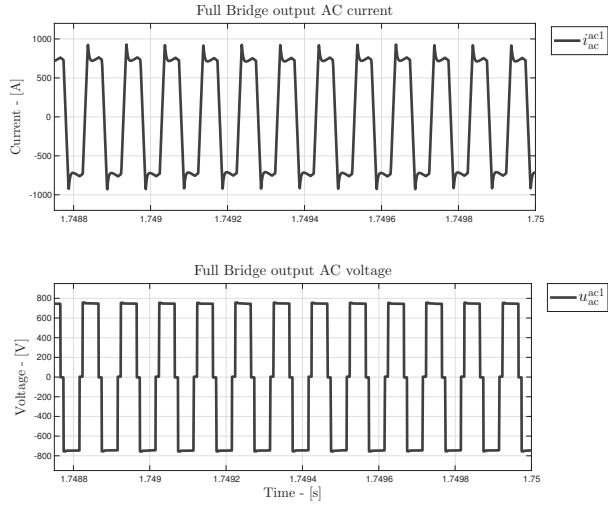


(a) Grid current waveform (top), and grid voltage waveform (bottom).

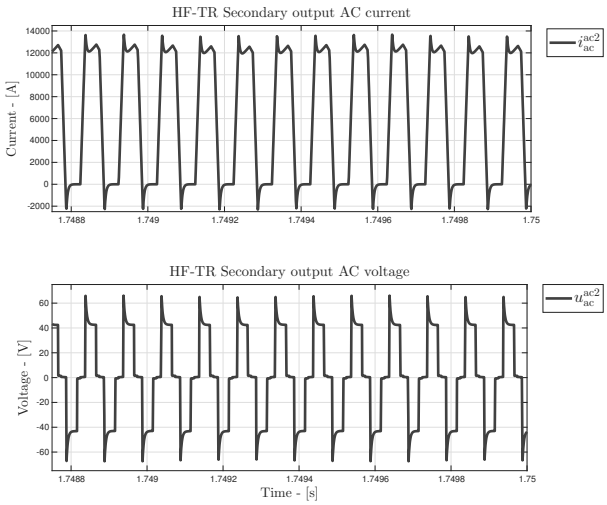


(b) AFE-Stack current waveform (top), and AFE-DClink Voltage waveform (bottom).

Figure 6: Grid and AFE stack quantities - case scenario NLC.

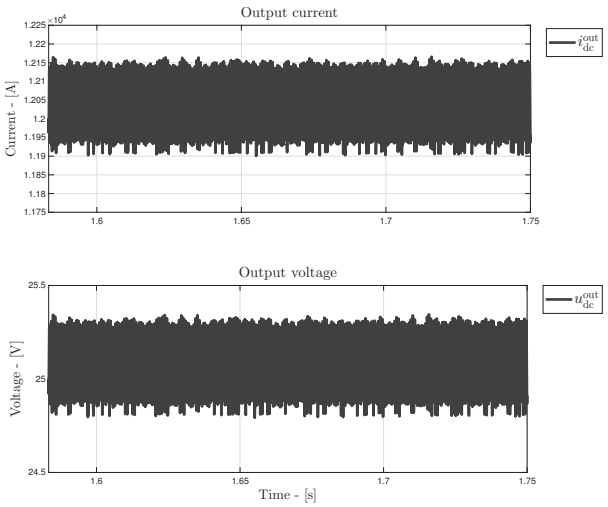


(a) Full Bridge output current (top), and output voltage (bottom).

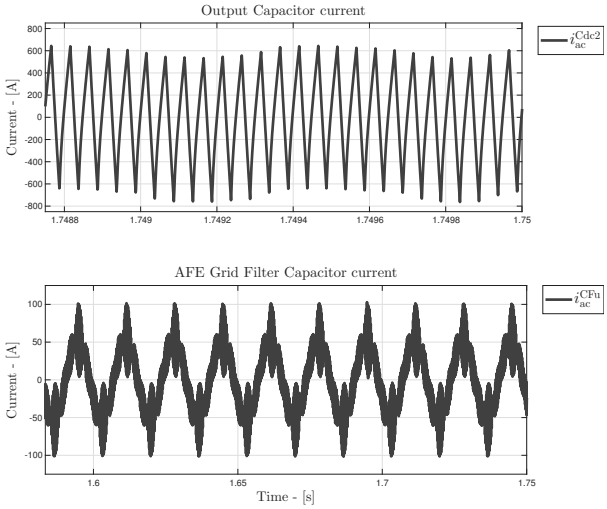


(b) HF-TR secondary current (top), and voltage (bottom).

Figure 7: Full Bridge Converter quantities - case scenario NLC.

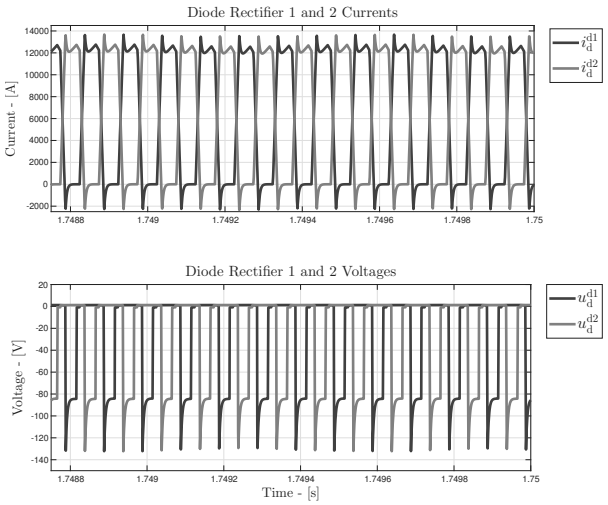


(a) Output current (top), and output voltage (bottom).

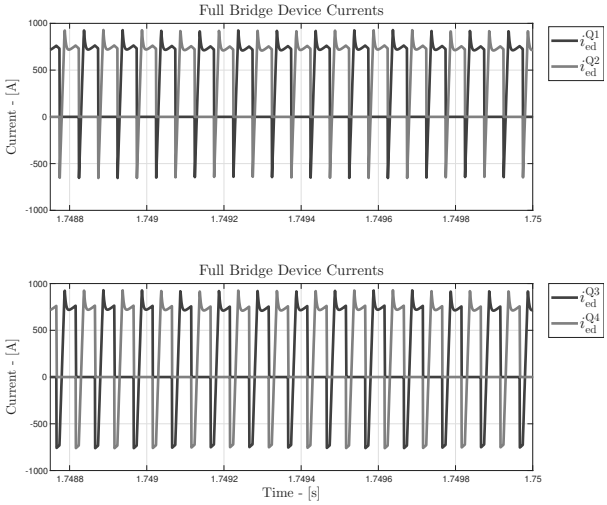


(b) Output filter capacitor current (top), and AFE grid filter capacitor current (bottom).

Figure 8: Output quantities and capacitor filter currents - case scenario NLC.

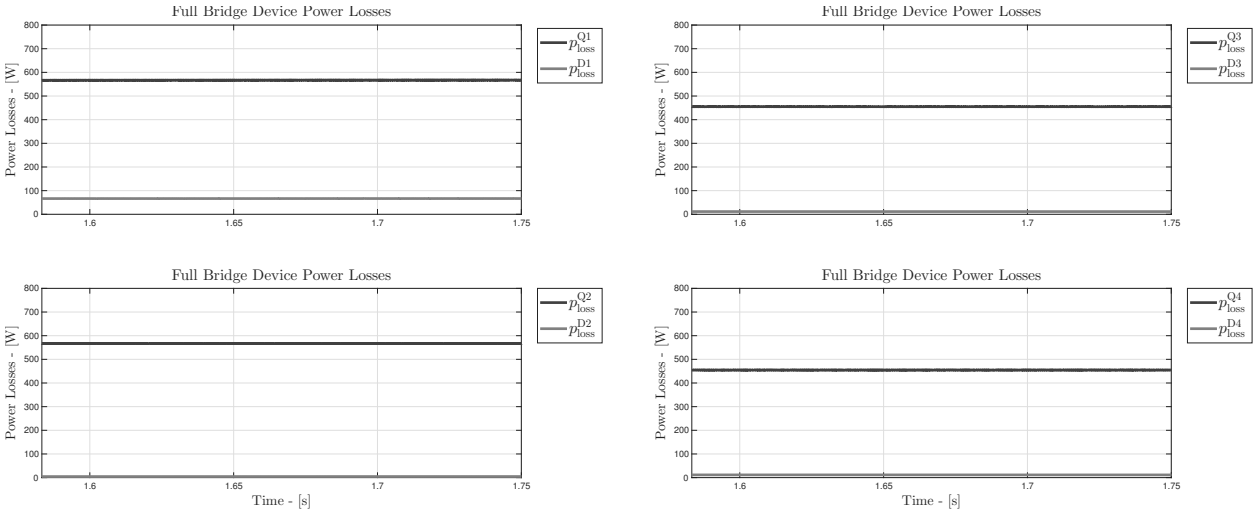


(a) Diode rectifiers current (top), and diode rectifiers voltage (bottom).



(b) Full Bridge devices current first leg (top), and second leg (bottom).

Figure 9: Full bridge devices current and diode rectifiers current - case scenario NLC.



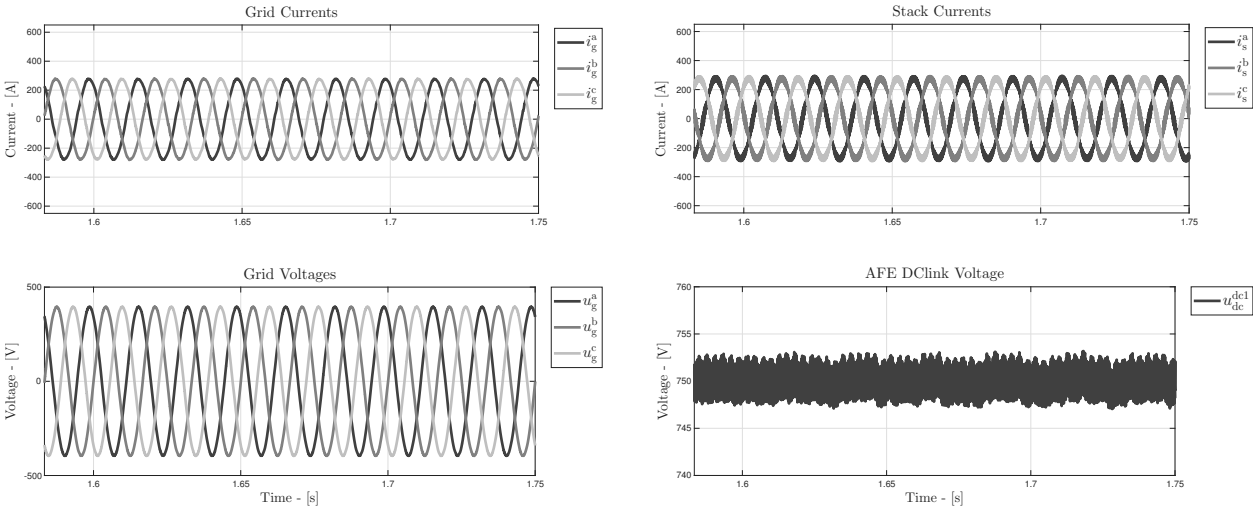
(a) Full Bridge device power losses (top), and (bottom).

(b) Full Bridge device power losses (top), and (bottom).

Figure 10: Full bridge device power losses - case scenario NLC.

5.2 Case scenario NLH

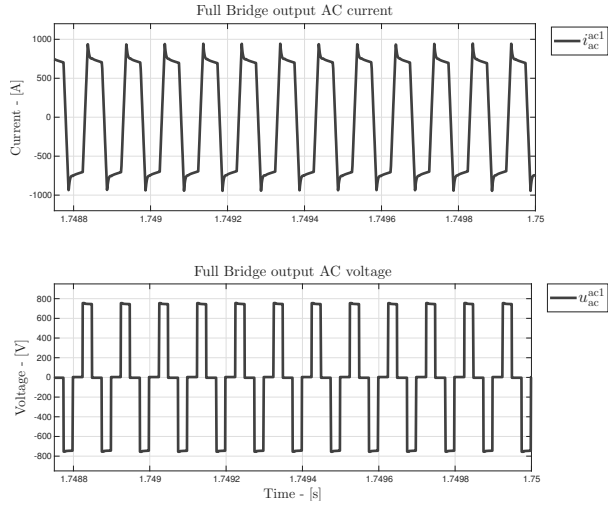
This subsection presents the simulation results for the NLH case. The simulations report the voltages and currents of the main passive and active components, as well as the corresponding losses.



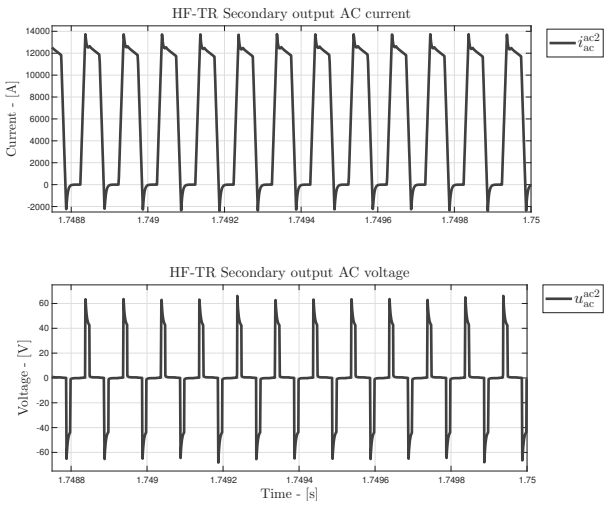
(a) Grid current waveform (top), and grid voltage waveform (bottom).

(b) AFE-Stack current waveform (top), and AFE-DClink Voltage waveform (bottom).

Figure 11: Grid and AFE stack quantities - case scenario NLH.

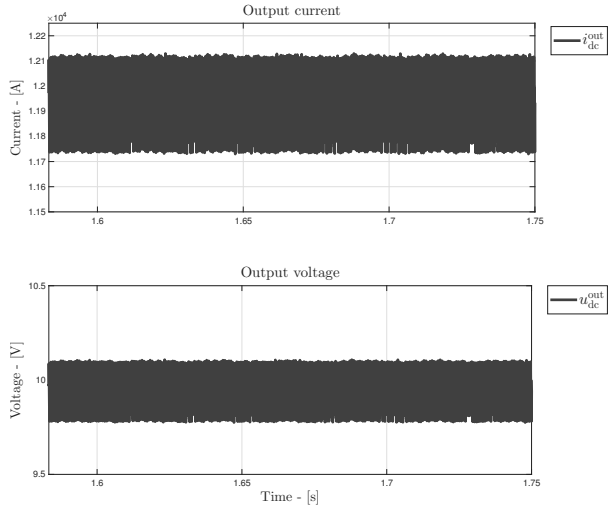


(a) Full Bridge output current (top), and output voltage (bottom).

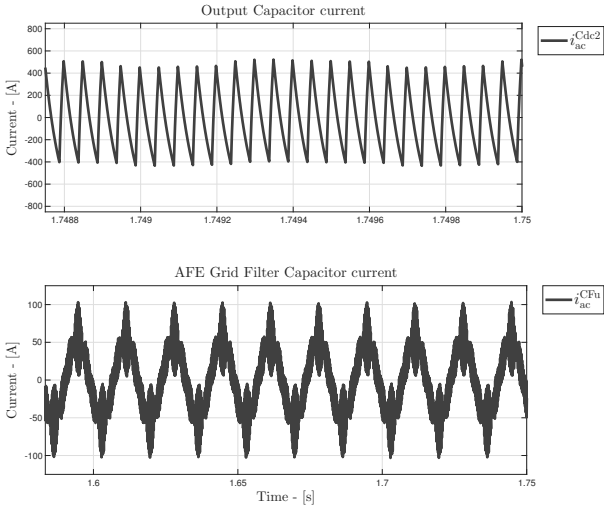


(b) HF-TR secondary current (top), and voltage (bottom).

Figure 12: Full Bridge Converter quantities - case scenario NLH.

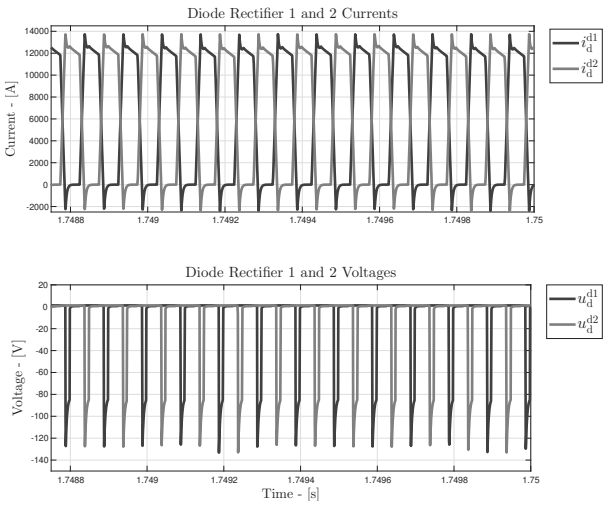


(a) Output current (top), and output voltage (bottom).

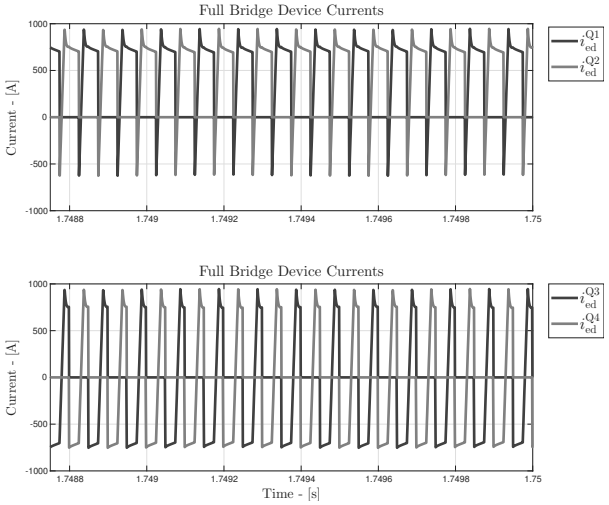


(b) Output filter capacitor current (top), and AFE grid filter capacitor current (bottom).

Figure 13: Output quantities and capacitor filter currents - case scenario NLH.

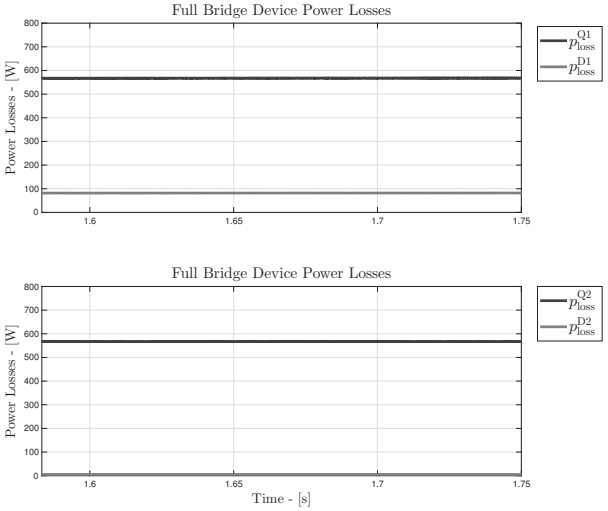


(a) Diode rectifier current (top), and (bottom).

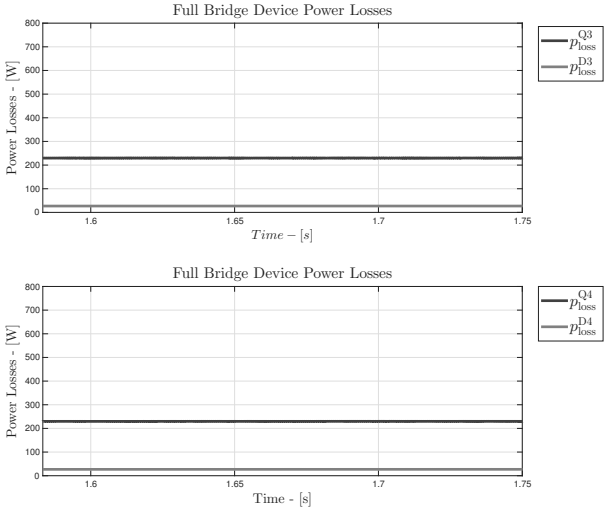


(b) Full Bridge devices current (top), and (bottom).

Figure 14: Full bridge devices current and diode rectifiers current - case scenario NLH.



(a) Full Bridge device power losses (top), and (bottom).



(b) Full Bridge device power losses (top), and (bottom).

Figure 15: Full bridge device power losses - case scenario NLH.